

management and handling rules.

Association sources blamed proliferation of unauthorised smelters and assemblers as well as lack of monitoring of dealers by state pollution control boards as the prime reason behind the poor response to the buyback schemes offered by the organised sector.

The industry needs to evolve with more people that know this segment technically.

Advances being made

As said earlier, compared to conventional flooded batteries, VRLA batteries do not need periodic topping-up. "However, routine maintenance in terms of checking of charging parameters, voltage reading, etc needs to be done for VRLA battery also. VRLA batteries are sensitive to charging parameters and depth of discharge, and if proper care and maintenance in terms of sizing and running the battery are not taken, there is a chance of failure," says Tarit Kumar Roy, general manager-institutional UPS, telecom, railways, solar, Exide Industries.

Exide Industries has recently launched a new-generation VRLA battery range branded as Powersafe NXT. Powersafe NXT has many unique features in terms of deep-discharge and fast recharge capabilities.

"In order to improve the lifetime of VRLA batteries and give more value to the customer, we have developed innovative plate chemistries for longer lifetime under Indian climatic conditions. We have also developed customised charging algorithms matching our battery technologies for the short-

est-timeframe, full-capacity charging of the batteries. This will be a huge advantage to the customer under Indian power conditions," says Sood of Regnart.

Rajesh Gupta, director, Okaya Power, reveals, "We use inter-cell weld of inner cell, which provides a very low-resistance current path.

We also use an extra-thick



Artheon batteries

At present, SMF batteries are 20 per cent costlier than conventional lead-acid batteries. However, through mass production, costs can be brought down



—Kunwer Sachdev, managing director, Su-Kam Power Systems

special alloy for very long life and preventing the water loss."

"There is a trend towards use of better methods to chemically process the battery (to prevent electrolyte loss), purer lead (to prevent plate corrosion), newer technologies in the charging equipment (to improve battery life) and thicker plates (to decrease charging time). By using thicker plates and increasing the charging current on our UPS, we are able to obtain the quickest charging time," says Aditya Arora, marketing head, Base Corporation.

Periodic maintenance of checking

torque of the terminal bolts, open-circuit voltage and internal resistance once in six months is recommended. There are many advances taking place in reducing corrosion and improving life through various combinations of the plate-paste mix. Battery charging is essentially being experimented with different types of battery chargers having improved capabilities like fuzzy logic.

How safe are these?

Since VRLA batteries are sealed, there is no leakage of electrolyte or gases.

The challenge for manufacturing in India largely centres around the availability of lead

—Biju Bruno, managing director, GreenVision Technologies



Relicell battery